SAECOIN — PROTOCOL REVIEW PLAN

Document: Protocol_Review_Plan.md
Project: SAECOIN
Ticker: SAC (Provisional)
Version: 1.0
Status: Planning
Purpose: Independent technical evaluation and validation of the SAECOIN blockchain protocol before implementation.

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1. PURPOSE

This document defines the research, mathematical verification, security analysis and independent review process required before implementing the SAECOIN blockchain protocol.

The primary objectives are:

- Select a suitable Proof of Work algorithm.
- Evaluate CPU and GPU mining compatibility.
- Assess ASIC resistance.
- Verify the complete monetary issuance schedule.
- Select secure cryptographic standards.
- Design a reliable difficulty adjustment mechanism.
- Validate transaction and block rules.
- Identify consensus vulnerabilities.
- Ensure deterministic behavior across independent nodes.

Important: No consensus-critical component should be treated as finalized until its specification has been reviewed and approved.

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2. REVIEW PRINCIPLES

All protocol decisions must follow these principles:

1. Security takes priority over convenience.
2. Prefer established cryptographic standards over custom cryptography.
3. Avoid unnecessary protocol complexity.
4. Consensus rules must be deterministic.
5. Independent nodes must reach identical validation results.
6. Monetary issuance must be mathematically verifiable.
7. Mining algorithms must be benchmarked using real hardware.
8. Every critical decision must have documented technical reasoning.
9. Security assumptions must be explicitly documented.

10. Changes to consensus rules require additional review.

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3. REVIEW TEAM AND RESPONSIBILITIES

Role| Responsibility
Project Owner| Approves project-level decisions
Blockchain Architect| Reviews protocol architecture
Rust Developer| Evaluates implementation feasibility
Cryptography Reviewer| Reviews cryptographic standards
Security Reviewer| Identifies security vulnerabilities
Mining Researcher| Evaluates PoW and mining performance
Independent Reviewer| Challenges assumptions and verifies results

One person may initially perform multiple roles during research, but independent review must be obtained before mainnet approval.

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4. PROOF OF WORK ALGORITHM REVIEW

Review ID: PR-001
Priority: CRITICAL

Objectives

Select a PoW algorithm appropriate for an independent blockchain supporting CPU and GPU mining.

Research Questions

- Which established algorithms are suitable?
- What memory requirements should mining have?
- How efficiently can CPUs execute the algorithm?
- How efficiently can GPUs execute the algorithm?
- What are the implementation requirements?
- How difficult would specialized hardware development be?
- What are the algorithm's known security limitations?

Candidate Evaluation

Research established algorithm families such as:

- SHA-256-based PoW.
- Scrypt.
- RandomX.
- Other memory-hard PoW designs.

These are research candidates, not approved SAECOIN selections.

Required Tests

- CPU benchmark.
- GPU benchmark.
- Memory consumption benchmark.
- Verification performance benchmark.
- Cross-platform compatibility test.
- Long-duration mining test.
- Independent algorithm security review.

Approval Requirements

- [ ] Candidate algorithms researched.
- [ ] Technical comparison completed.
- [ ] CPU benchmarks completed.
- [ ] GPU benchmarks completed.
- [ ] Security assumptions documented.
- [ ] ASIC resistance limitations documented.
- [ ] Final algorithm selected.
- [ ] Independent review completed.

Deliverable: Final PoW Algorithm Decision.

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5. DIFFICULTY ADJUSTMENT REVIEW

Review ID: PR-002
Priority: CRITICAL

Objectives

Design a difficulty adjustment mechanism that maintains stable block production while resisting manipulation.

Research Questions

- How frequently should difficulty change?
- What happens when mining power increases suddenly?
- What happens when mining power decreases suddenly?
- Can timestamps be manipulated?
- Can miners manipulate historical block times?
- How should the system behave during low hash-rate periods?

Required Simulations

- Stable hash rate.
- Sudden hash-rate increase.
- Sudden hash-rate decrease.
- Temporary mining interruption.
- Unstable timestamp behavior.
- Short-term mining power manipulation.
- Network partition and recovery.

Approval Requirements

- [ ] Candidate formulas researched.
- [ ] Mathematical model documented.
- [ ] Simulation completed.
- [ ] Timestamp attack analysis completed.
- [ ] Difficulty manipulation analysis completed.
- [ ] Edge cases documented.
- [ ] Final algorithm approved.

Deliverable: Final Difficulty Adjustment Specification.

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6. MONETARY ISSUANCE REVIEW

Review ID: PR-003
Priority: CRITICAL

Current Proposal

Parameter| Proposed Value
Maximum Supply| 21,000,000 SAC
Initial Reward| 50 SAC
Halving Interval| 210,000 Blocks
Decimal Precision| 8
Premine| 0 SAC

Objectives

Verify that the monetary rules enforce the intended supply limit.

Mathematical Verification

The review must calculate:

- Initial block subsidy.
- Subsidy after every halving.
- Total coins created in each reward era.

- Total issuance across all eras.
- Effects of integer rounding.
- Final subsidy behavior.
- Maximum possible supply.

The implementation must use integer arithmetic rather than floating-point calculations for monetary values.

Required Tests

- [ ] Simulate the full issuance schedule.
- [ ] Verify every halving boundary.
- [ ] Test reward calculation at boundary blocks.
- [ ] Test smallest-unit rounding.
- [ ] Test subsidy after repeated halvings.
- [ ] Test zero-subsidy behavior.
- [ ] Verify coinbase validation.
- [ ] Attempt unauthorized coin creation.
- [ ] Independently verify maximum supply.

Deliverable: Verified Monetary Issuance Specification.

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7. CRYPTOGRAPHIC STANDARDS REVIEW

Review ID: PR-004
Priority: CRITICAL

Objectives

Select secure and interoperable cryptographic standards for transactions, wallets and block validation.

Components to Review

Hash Functions

Evaluate:

- Security properties.
- Collision resistance.
- Preimage resistance.
- Performance.
- Availability of reliable implementations.

Digital Signatures

Evaluate:

- Signature security.
- Public key format.
- Signature size.
- Verification performance.
- Wallet compatibility.
- Implementation maturity.

Key Generation

Evaluate:

- Cryptographically secure randomness.
- Private key generation.
- Public key derivation.
- Key storage.
- Key recovery.

Wallet Encryption

Evaluate:

- Password-based key derivation.
- Authenticated encryption.
- Salt generation.
- Protection against brute-force attacks.
- Secure backup procedures.

Approval Requirements

- [ ] Hash standard selected.
- [ ] Signature standard selected.
- [ ] Key generation standard selected.
- [ ] Public key encoding finalized.
- [ ] Wallet recovery standard selected.
- [ ] Wallet encryption design reviewed.
- [ ] Cryptographic dependencies audited.
- [ ] Independent cryptographic review completed.

Rule: Do not invent custom cryptographic algorithms for SAECOIN.

Deliverable: Cryptographic Standards Specification.

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8. TRANSACTION MODEL REVIEW

Review ID: PR-005
Priority: CRITICAL

Current Proposal: UTXO.

Objectives

Ensure transaction validation is secure, deterministic and resistant to unauthorized spending.

Review Areas

- Transaction input validation.
- Transaction output validation.
- Signature verification.
- Double-spend prevention.
- Transaction serialization.
- Transaction identifiers.
- Fee calculation.
- Coinbase validation.
- UTXO creation and spending.
- Integer overflow prevention.
- Transaction malleability considerations.

Required Tests

- [ ] Valid transaction tests.
- [ ] Invalid signature tests.
- [ ] Double-spend tests.
- [ ] Invalid input tests.
- [ ] Invalid output tests.
- [ ] Integer overflow tests.
- [ ] Malformed serialization tests.
- [ ] Coinbase reward tests.
- [ ] Reorganization-related UTXO tests.

Deliverable: Final Transaction Validation Specification.

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9. BLOCK AND CHAIN VALIDATION REVIEW

Review ID: PR-006
Priority: CRITICAL

Review Areas

- Block header structure.

- Block hash calculation.
- Merkle root calculation.
- Proof of work verification.
- Timestamp validation.
- Difficulty validation.
- Block reward validation.
- Transaction ordering.
- Maximum block size or weight.
- Chain selection.
- Chain reorganization.
- Genesis block validation.

Required Tests

- [ ] Valid block validation.
- [ ] Invalid proof of work rejection.
- [ ] Invalid timestamp rejection.
- [ ] Incorrect Merkle root rejection.
- [ ] Excessive reward rejection.
- [ ] Invalid transaction rejection.
- [ ] Competing chain tests.
- [ ] Deep reorganization tests.
- [ ] Genesis block consistency tests.

Deliverable: Final Block and Chain Validation Specification.

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10. NETWORK SECURITY REVIEW

Review ID: PR-007
Priority: HIGH

Threats to Evaluate

- Sybil attacks.
- Denial-of-service attacks.
- Malformed network messages.
- Peer flooding.
- Eclipse attacks.
- Network partition.
- Block propagation delays.
- Transaction flooding.
- Malicious peer behavior.

Required Tests

- [ ] Peer connection stress tests.

- [ ] Malformed message tests.
- [ ] Connection limit tests.
- [ ] Rate limit tests.
- [ ] Network partition simulation.
- [ ] Recovery and reconnection tests.
- [ ] Block propagation tests.
- [ ] Independent node synchronization tests.

Deliverable: P2P Security Review Report.

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11. ASIC RESISTANCE EVALUATION

Review ID: PR-008
Priority: HIGH

Objectives

Evaluate whether the selected PoW algorithm meaningfully reduces specialized hardware advantages.

Evaluation Criteria

- Memory requirements.
- Memory bandwidth requirements.
- CPU performance.
- GPU performance.
- Specialized hardware feasibility.
- Energy efficiency differences.
- Hardware accessibility.
- Long-term algorithm adaptability.

Important Considerations

No algorithm can be guaranteed to remain ASIC-resistant forever.

The review must distinguish between:

- CPU-friendly.
- GPU-friendly.
- Memory-hard.
- ASIC-resistant by design objective.
- Demonstrably resistant under a defined benchmark.

Deliverable: Mining Hardware Resistance Assessment.

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12. GENESIS BLOCK REVIEW

Review ID: PR-009
Priority: CRITICAL

Objectives

Ensure every official SAECOIN node recognizes exactly the same genesis block.

Review Requirements

- [ ] Genesis configuration finalized.
- [ ] Genesis timestamp verified.
- [ ] Genesis transaction verified.
- [ ] Genesis Merkle root verified.
- [ ] Genesis hash independently calculated.
- [ ] Genesis proof of work verified.
- [ ] Genesis configuration stored in version control.
- [ ] Independent implementation verification completed.

Deliverable: Verified Genesis Block Manifest.

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13. DATABASE AND STATE REVIEW

Review ID: PR-010
Priority: HIGH

Objectives

Evaluate the correctness and reliability of blockchain state storage.

Review Areas

- UTXO storage.
- Block storage.
- Transaction indexing.
- Chainstate consistency.
- Database transactions.
- Crash recovery.
- Corruption recovery.
- Backup and restoration.
- Reorganization handling.
- Disk usage growth.

RocksDB remains a proposed option until compatibility, reliability and performance are evaluated.

Deliverable: Database Architecture Review.

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14. SECURITY THREAT MODEL

Review ID: PR-011
Priority: HIGH

Identify:

- Assets requiring protection.
- Potential attackers.
- Attack capabilities.
- Network-level threats.
- Consensus-level threats.
- Wallet-level threats.
- Infrastructure-level threats.
- Supply manipulation risks.
- Key compromise risks.

For every identified threat, document:

- Attack scenario.
- Potential impact.
- Likelihood.
- Mitigation.
- Residual risk.
- Required testing.

Deliverable: Threat_Model.md.

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15. INDEPENDENT REVIEW PROCESS

The following components require independent review:

Component| Independent Review
PoW Algorithm| Required
Difficulty Adjustment| Required
Monetary Issuance| Required
Cryptography| Required
Transaction Validation| Required
Block Validation| Required

Genesis Block| Required
P2P Security| Required
Wallet Security| Required
Database Recovery| Recommended

Independent reviewers should receive:

- Relevant specification documents.
- Mathematical calculations.
- Source code when available.
- Test results.
- Known limitations.
- Unresolved questions.

Review findings must be documented and tracked until resolved or formally accepted with justification.

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16. REVIEW APPROVAL PROCESS

Each review must follow these stages:

Stage 1 — Research

Collect technical information and evaluate established solutions.

Stage 2 — Proposal

Document candidate solutions and explain their advantages and disadvantages.

Stage 3 — Internal Verification

Perform mathematical checks, simulations and implementation feasibility analysis.

Stage 4 — Independent Review

Obtain review from someone who did not make the original design decision.

Stage 5 — Approval

Record the selected solution and supporting evidence.

Stage 6 — Documentation Update

Update all related SAECOIN specifications.

Stage 7 — Implementation Authorization

Only after the relevant requirements are approved should implementation of that component begin.

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17. REVIEW TRACKING TABLE

Review ID| Subject| Priority| Status
PR-001| PoW Algorithm| Critical| Open
PR-002| Difficulty Adjustment| Critical| Open
PR-003| Monetary Issuance| Critical| Open
PR-004| Cryptographic Standards| Critical| Open
PR-005| Transaction Model| Critical| Open
PR-006| Block and Chain Validation| Critical| Open
PR-007| Network Security| High| Open
PR-008| ASIC Resistance| High| Open
PR-009| Genesis Block| Critical| Open
PR-010| Database and State| High| Open
PR-011| Threat Model| High| Open

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18. FINAL PROTOCOL APPROVAL

Before consensus-critical implementation is authorized:

- [ ] PoW algorithm approved.
- [ ] Difficulty adjustment approved.
- [ ] Monetary issuance verified.
- [ ] Cryptographic standards approved.
- [ ] Transaction rules approved.
- [ ] Block validation rules approved.
- [ ] Chain selection rules approved.
- [ ] Genesis specification approved.
- [ ] Network parameters approved.
- [ ] Threat model completed.
- [ ] Independent review completed.
- [ ] Critical findings resolved.
- [ ] Related documentation updated.

Approval Record

Protocol Version: _____

Review Completion Date: _____

Technical Reviewer: _____

Independent Reviewer: _____

Project Approval: _____

Decision:

- [ ] Approved for implementation.
- [ ] Approved with documented conditions.
- [ ] Requires additional research.
- [ ] Rejected.

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19. FINAL PRINCIPLE

SAECOIN must be designed around verifiable technical rules rather than assumptions.

A working mining demonstration does not prove that a blockchain is secure.

A successful software build does not prove that consensus is correct.

A mathematically verified supply schedule does not prove that the implementation enforces it correctly.

Protocol research, independent review, implementation testing and public testnet validation are separate requirements. All are necessary before mainnet launch.